

JESTIN THOMAS

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Professional Summary

MSc Data Science (CHRIST University, 2026), BSc Statistics, **6 months** AI/ML engineering experience. IEEE-published researcher and Government of India **Design Patent** holder. Proficient in **ML/DL, NLP, LLMs, RAG, AI Agents, Generative AI, EDA, predictive analytics, hypothesis testing, A/B testing, regression modeling, sampling and representation, and data-driven decision making**. Delivered production-grade systems across **AV cybersecurity, market intelligence, scientific classification, and retail BI**; builds reproducible pipelines following MLOps best practices; proficient with **Git, AWS, and Google Cloud**.

Skills

Programming: Python, R, SQL, Java, JavaScript, Pandas, NumPy, Scikit-learn, TensorFlow, Keras

ML and AI: Classification, Regression, Clustering, Random Forest, XGBoost, CNN, LSTM, GRU, Transformers, LLMs, Prompt Engineering, RAG, AI Agents, LangChain, GANs, VAE, Diffusion Models, GNN, Transfer Learning, A/B Testing, Time Series

Data Science: EDA, Feature Engineering, Missing Data Imputation, Sampling and Representation, Hypothesis Testing, Trend Analysis, Predictive Analytics, Statistical Modeling, Data Integration

Visualization and BI: Power BI, Tableau, Matplotlib, Seaborn, Plotly, Streamlit, KPI Reporting, Dashboard Development

Cloud and Tools: AWS, Google Cloud, PostgreSQL, MySQL, Git, Flask, FastAPI, Docker, Streamlit Cloud, MLOps, Excel

Professional Experience

AI/ML Engineer Intern | TechZia

Jan 2026 – Jun 2026

Autonomous Vehicle Cybersecurity Agent

- Designed **CNN + LSTM** real-time anomaly detection pipeline on AV telemetry (IMU, CAN bus, GPS) in **ROS 2 Humble**, monitoring **3 concurrent sensor streams** with sub-100ms inference latency
- Built **DQN-based multi-agent reinforcement learning** system with 3 specialized RL agents (sensor security, vehicle behavior, network cybersecurity) coordinated by a hierarchical global decision agent; reduced pipeline failure rate by identifying and resolving production quality issues
- Preprocessed **comma.ai AV dataset** using vectorized NumPy pipelines with bias reduction and dimensionality reduction; researched CNN-Transformer and GNN-based multi-sensor fusion architectures for network intrusion detection

Projects

Zomato Market Intelligence Platform | Python, Pandas, Scikit-learn, Streamlit, Plotly

GitHub | [Live Demo](#)

- Analyzed **12,000+ restaurants across 90+ locations** in Bengaluru; performed EDA, feature engineering, sentiment analysis, customer behavior analysis, and market segmentation to extract actionable business insights
- Trained **Random Forest Regressor** for restaurant rating prediction and applied **K-Means Clustering** for market segmentation; engineered 4 custom business metrics: **Success Score, Cuisine Popularity Index, Cost Segmentation, and Location Performance Analytics**
- Built **Market Expansion Advisor** recommending profitable locations based on cuisine demand, budget, and competition density; deployed full-stack **Streamlit** application on Streamlit Cloud using Git version control and industry-standard MLOps project structure

Solar Flare Classification | Python, DenseNet121, GRU, Attention Mechanism, TensorFlow.js, React

[GitHub](#)

- Developed **DenseNet121 + GRU + Attention** hybrid model trained on **8,000+ satellite image sequences**, achieving **75% accuracy** with 58% M-class and 36% X-class recall on rare solar events; addressed class imbalance using sampling and indirect estimation techniques
- Deployed model client-side using **TensorFlow.js** in a React application (zero backend infrastructure); research methodology published at IEEE ICVADV-2026

Retail Profitability Intelligence Dashboard | Python, SQL, Apache Spark, Power BI, Tableau

[GitHub](#)

- Processed retail transaction data using **Apache Spark and SQL aggregations**; performed EDA, missing data imputation, and dimensionality reduction; engineered **Profit Margin** and **Risk Score** KPIs for discount impact and trend analysis
- Delivered interactive **Power BI and Tableau dashboards** with KPI cards (Sales, Profit, Orders, Average Margin), regional breakdowns, and dynamic filters; insights directly supported pricing and inventory decisions

Education

Master of Science in Data Science | CHRIST (Deemed to be University), Bengaluru

2024 – 2026

Bachelor of Science in Statistics | Sree Sankara College, Kalady (MG University)

2021 – 2024

Research and Intellectual Property

IEEE Conference Paper — ICVADV-2026, [IEEE Xplore](#)

Feb 2026

Synthetic Data Augmentation for Robust Solar Flare Classification: A Comparative Analysis of Conditional DCGAN, VAE, and Diffusion Models

- Benchmarked 3 generative deep learning architectures (Conditional DCGAN, VAE, Diffusion Models) for synthetic data augmentation; demonstrated measurable improvement in classifier robustness on imbalanced astronomical datasets

Design Patent — Government of India — *Mobile Cooler* — Class 23-04 — **No. 443142-001** — **Jan 2025**